

Thinking Machines, Human Choices

Power and Responsibility in AI Design and
Deployment



About Sarah (she/her)

- Majored in Economics, transitioned into tech in my early 20s
- Worked professionally as a software engineer for 10 years before grad school (working on collaborative decision-making systems for state and local governments)
- Went to graduate school to research the societal applications and impacts of tech
- My current research & professional interests include:
 - Examining the societal impacts of emerging technologies & business models
 - Building educational tools and learning experience to help novices learn about and understand ideas in data science and computing

Let's Talk!

Break into groups of 3-4. Introduce yourself (go around):

1. Name, pronouns, something about you – fun fact, what you did over summer break, pets, hobbies, morning person v. night person.
2. Why are you here? What do you want to learn?
3. What's your major?
4. What's a recent story you've heard about AI – either in the news, on social media, from friends, family, or somewhere else?

Today's Questions

- What the heck is AI?
- What is AI supposed to help us do?
- Where is AI already shaping the world?
- What do we value about it?
- What worries us?
- What should we learn to be able to decide whether and how “we” want to use it – both as individuals and as a society?

When You Hear “AI” What Comes To Mind?

- Terminator
- Generative
- Llms
- Robots
- Confusing
- Possible demise
- Orwell
- Existential threat
- Paperclips
- China
- UBI
- Water
- SciFi
- Oppression
- Automation
- Convenience
- Work ‘assistance’

What do we like? What don't we like?

What we like

- Your thoughts here...

What we don't like

- Your thoughts here...

Why is AI is a public issue? Why is everyone talking about it?

- Integral part of our lives.
- Learning
- Work
- Who benefits? Who bears the costs?
- Environmental
- Trains off of public data?
- Consent and theft
- Lack of human connection
- Her
- Promoting laziness
- enables a lack of creativity
- Privacy – and opacity – and surveillance.
- Media literacy – not engaging w/source material.
- Dead internet theory
- Opt in / opt out
- Facial recognition, etc.
- AI artists – only possible b/c of mass theft?

Where is AI already showing up?

Can you come up with some examples of how AI is being used in each of these domains / sectors?

- school
- work
- healthcare
- entertainment
- news/social media
- hiring
- government
- creative work
- relationships
- transportation

What Is AI Being Used to Help Someone Do?

- predict
- recommend
- classify
- generate
- summarize
- monitor
- optimize
- decide
- persuade
- Obtaining (by ill means)?

And who is the “someone”?

AI debates are often about values.

Value: A principle or standard of behaviour; a judgment about what is important in life.

autonomy
efficiency
accuracy
fairness
privacy
dignity
convenience

safety
creativity
expertise
accountability
personalization
human
connection
explainability

These values may all seem important and relatively uncontroversial on their own. But in practice, they often come into conflict!

- When values conflict, who gets to decide which value matters more?
- Who receives the benefits of that decision – and who bears the costs?

Why is AI is a public issue?

AI is not just a tool! It is part of a system involving:

**technology + people + institutions +
values + decisions + consequences**

Example: AI in Hiring

Technical question:

- What is the system trying to predict?
- How does it make that prediction?
- How accurate is it?

Social questions:

- What counts as “good”?
- What data are used?
- Who is excluded?
- Who chooses the threshold?
- Who bears the cost of error?
- Should hiring work this way?

How do we deconstruct all of these
issues, questions, claims, news
stories?!

Different Kinds of Claims

When we talk about AI, we often mix together descriptive / predictive, normative, and political / institutional claims

Descriptive / Predictive Claims

Claims about what is, what is happening, or what a system can do.

- “Employers are using AI to screen job applicants.”
- “Some facial-recognition systems perform differently across demographic groups.”
- “AI is becoming sentient.”

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Political / Institutional Claims

Claims about **who has power** or authority, or who gets to decide

- “Employers decide which criteria an AI hiring system uses to evaluate applicants.”
- “A few tech companies control access to many of the most capable AI models.”

Questions to ask:

- What evidence supports the **descriptive** claim?
- What values or assumptions shape the **normative** claim?
- What actors, institutions, or power structures shape the **political / institutional** claim?

Activity: What Kind of Claim Is This?

Split into groups of 4 and complete [Activity 1. What Kind of Claim Is This?](#)

You have 15 minutes.

Share Out

Your thoughts here...

Course Format & Policies

What Kind of Course Is This?

This is not a programming course. We will study:

- Human learning and judgment
- How AI works
- Data and prediction
- Institutions and domains
- Power and political economy
- Governance and future technologies

How We'll Work Together

Seminar expectations:

- Come prepared
- Bring your laptop
- Participate actively
- Ask questions
- Disagree respectfully
- Connect ideas to your field
- Distinguish evidence from speculation

01

Humans,
Intelligence,
and
Judgment



02

Machines,
Data, and AI



03

A Toolkit for
Under-
standing AI
in Society



04

AI Enters
the World



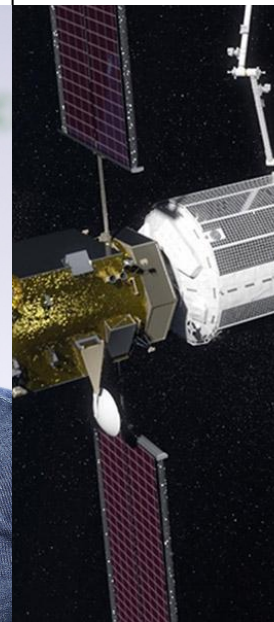
05

Patterns
Across
Domains:
AI, Power,
and Political
Economy



06

Governing
Futures
That Do Not
Yet Exist



07

Synthesis:
Professional
and Civic
Responsi-
bility



AI Use in This Course

- We will use AI in some activities.
- We will restrict AI in others.
- Especially: reading responses should reflect your own encounter with the material.

NSF Project: Ethical, Legal, & Societal Implications of Emerging AI Hardware

I am also part of a team – in collaboration with Northwestern University – to investigate the implications of some new brain-inspired AI hardware, to make the next generation of AI technologies (more on that later in the semester).

You will be a part of this work in helping the field think through some of the broader impacts of AI as their capabilities become more sophisticated.

What to Expect / Grading

Assignment	Percentage of Grade
Reading Preparation & Participation	15%
Homework & Labs	35%
Development Activities	25%
Applied AI & Society Final Project	25%

Attendance Policy

This is a discussion-based senior seminar, so your presence matters.

- You may miss 2 classes with no penalty – no questions asked.
- 3rd absence: final course grade lowered by half a letter.
- Each additional absence: another half-letter reduction.

If ongoing circumstances may affect your attendance, talk with me early.

- Please save your two no-penalty absences for illness, emergencies, or other situations when you genuinely need them.

Late Work Policy

- **Reading responses** are intended to prepare you for class discussion and should be completed before the associated class meeting. Late reading responses will not receive credit.
- **Lab assignments** submitted within one week of the deadline will receive a 20% late penalty. After one week, late work will not be accepted.
- **Time-sensitive assignments:** Some assignments involve presentations, collaborative activities, or other time-sensitive work and may have deadlines that cannot be extended in the same way. Any such expectations will be identified in the assignment instructions.

Homework For Thursday

Readings:

1. Saplakoglu, Y. (2026, April 24). A new type of neuroplasticity rewires the brain after a single experience. Quanta Magazine. [Quanta Magazine](#)
2. Vignola, N. (2024). A neuroscientist's guide to reclaiming your brain. Big Think. [YouTube](#) (8 minutes)
3. Sentis. (2012). Neuroplasticity. [YouTube](#) (20 minutes)

Discussion Forum (15 mins): Respond individually [on Canvas](#):

1. One use of AI I find valuable is...
2. One use of AI that gives me pause is...
3. One question I hope this course helps me answer is...